Transposable Element Sequence-Divergence Profiles in *Trichinella spiralis* and *Trichinella britovi*

Do contrasting host associations coincide with different TE histories in *Trichinella spiralis* and *Trichinella britovi*?

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What is Trichinella?

Trichinella is a genus of parasitic nematodes (roundworms) that infect vertebrates. Unlike many parasitic worms, Trichinella completes its entire life cycle within a single host species lineage and does not require a free-living environmental stage. [1]



Key Biological Features

1. Intracellular lifestyle: Larvae invade skeletal muscle cells and transform them into specialized "nurse cells" that support long-term survival.
2. Direct life cycle: The same host acts as both intermediate and definitive host.
3. Viviparity: Adult females release live larvae rather than laying eggs.
4. Broad host range: Different species infect mammals, birds, and reptiles.
5. Trophic transmission: Transmission occurs through predation or scavenging when infected muscle tissue is eaten. [2]

[1] Pozio, E., La Rosa, G., Murrell, K. D., & Lichtenfels, J. R. (1992). Taxonomic revision of the genus *Trichinella*. *The Journal of Parasitology, 78*(4), 654–659. https://doi.org/10.2307/3283540
[2] Mitic, I., Vasilev, S., & Gruden-Movsesijan, A. (2026). Trichinellosis: A zoonosis that still requires vigilance. *PLOS Neglected Tropical Diseases, 20*(1), e0013944. https://doi.org/10.1371/journal.pntd.0013944

T. spiralis and *T. britovi*: related species, different ecological associations

Feature	*T. spiralis*	*T. britovi*
Evolutionary relationship	Encapsulated *Trichinella* clade [1]	Encapsulated *Trichinella* clade [1]
Main host associations	Domestic pigs, wild boar, and rodents [2,3]	Wild carnivores, especially foxes and wolves; also wild boar and pigs [2,3]
Transmission cycles	Strong domestic association; also wildlife [2]	Mainly wildlife; can enter domestic cycles [2]
Geographic distribution	Widespread across Europe, Asia, and the Americas [3]	Temperate Europe, western/central Asia, and northern/western Africa [3]

[1] Korhonen, P. K., Pozio, E., La Rosa, G., et al. (2016). Phylogenomic and biogeographic reconstruction of the *Trichinella* complex. *Nature Communications, 7*, 10513. https://doi.org/10.1038/ncomms10513
[2] Pozio, E., Rinaldi, L., Marucci, G., Musella, V., Galati, F., Cringoli, G., Boireau, P., & La Rosa, G. (2009). Hosts and habitats of *Trichinella spiralis* and *Trichinella britovi* in Europe. *International Journal for Parasitology, 39*(1), 71–79. https://doi.org/10.1016/j.ijpara.2008.06.006
[3] World Organisation for Animal Health. (2023). Trichinellosis (infection with *Trichinella* spp.). In *Manual of Diagnostic Tests and Vaccines for Terrestrial Animals* (Chapter 3.1.22).

What are transposable elements?

Transposable elements (TEs) are DNA sequences capable of moving or generating new copies at different locations within a genome [3].

Broad class	How they move	Examples
Class I: Retrotransposons	Use an RNA intermediate to generate new DNA copies	LTR elements, LINEs, SINEs
Class II: DNA transposons	Move without an RNA intermediate; many use a "cut-and-paste" mechanism	Tc1/Mariner, hAT

Not all repetitive DNA is a TE: Simple repeats, such as repeated short DNA motifs, and low-complexity sequences are separate annotation categories [4].

An annotation is not proof of activity; An annotated TE region may be a fragment or an inactive remnant, rather than a complete, mobile element [4].

[3] Wicker, T., et al. (2007). A unified classification system for eukaryotic transposable elements. *Nature Reviews Genetics, 8*, 973–982. https://doi.org/10.1038/nrg2165
[4] RepeatMasker. (n.d.). *RepeatMasker web server documentation*. https://repeatmasker.org/webrepeatmaskerhelp.html

Why compare *T. spiralis* and *T. britovi*?

- **Shared ancestry:** as close relatives, the two species inherited most of their genome, including old TEs, from a common ancestor. Differences that arose after the split are therefore easier to detect (Korhonen et al. 2016).
- **Contrasting ecologies:** *T. spiralis* is closely tied to domestic pigs, a cycle established with pig domestication within the last ~10,000 years, while *T. britovi* circulates mainly in wildlife (Pozio et al. 2009).
- **TEs as a source of adaptive variation:** TE insertions have contributed to adaptation in diverse organisms. In plant-parasitic nematodes, TE activity is lineage-specific, and some TEs carry genes that may support parasitism (Dayi, 2024).
- **Shared public-health importance:** both species cause human trichinellosis.

Comparing TE divergence profiles can reveal patterns consistent with different TE histories. These data cannot date insertions to pig domestication or show that TEs caused adaptation.

[1] Korhonen, P. K., Pozio, E., La Rosa, G., et al. (2016). Phylogenomic and biogeographic reconstruction of the *Trichinella* complex. *Nature Communications, 7*, 10513. https://doi.org/10.1038/ncomms10513
[2] Pozio, E., La Rosa, G., Murrell, K. D., & Lichtenfels, J. R. (1992). Taxonomic revision of the genus *Trichinella*. *The Journal of Parasitology, 78*(4), 654–659. https://doi.org/10.2307/3283540
[3] Dayi, M. Diversity and evolution of transposable elements in the plant-parasitic nematodes. BMC Genomics 25, 511 (2024). https://doi.org/10.1186/s12864-024-10435-7

Existing research and ecological relevance

- **Variation within Trichinella:** Comparative genomics revealed differences in repetitive DNA content among *Trichinella* taxa, motivating comparisons of their TE landscapes (Korhonen et al., 2016).
- **Evidence from other nematodes:** Dayi (2024) found differences in TE composition and inferred age profiles among plant-parasitic nematodes. Some Gypsy retrotransposons encoded proteases, suggesting a possible connection between TEs and functions involved in parasitism.
- **Connection to our comparison:** *T. spiralis* and *T. britovi* have different host associations and transmission cycles (Pozio et al., 2009). This motivates exploring whether they also differ in TE divergence profiles.

Ecological differences provide context, but our analysis cannot establish that host associations caused differences in TE history.

[5] Korhonen, P. K., Pozio, E., La Rosa, G., et al. (2016). Phylogenomic and biogeographic reconstruction of the *Trichinella*complex. *Nature Communications, 7*, 10513. https://doi.org/10.1038/ncomms10513
[6] Dayi, M. (2024). Diversity and evolution of transposable elements in the plant-parasitic nematodes. BMC Genomics, 25, 511. https://doi.org/10.1186/s12864-024-10435-7

What will we compare?

- **Data:** RepeatMasker annotations from the *T. spiralis* and *T. britovi* genome assemblies.
- **Unit of comparison:** Individual annotated TE regions, grouped by their matching class. Regions may represent complete elements or fragments.
- **Measurement:** Percent sequence divergence from the corresponding TE consensus sequence, a reference representing that TE family.
- **Interpretation:** Lower divergence generally suggests relatively younger sequences, while higher divergence suggests older sequences. This is an approximate age indicator, not a direct date.
- **Scope:** Exclude designated non-TE categories and annotations labelled "Unknown."

Sequence divergence alone cannot confirm current TE activity or determine whether an insertion occurred before or after the species split.

Research question and objective

Research question

Do TE sequence-divergence distributions differ between *Trichinella spiralis* and *Trichinella britovi* within matching classes?

Objective

Identify which matching classes show differences between species and explore what these patterns suggest about their TE histories.

Approach

Compare percent-divergence distributions using Wilcoxon rank-sum tests within matching classes, with Benjamini–Hochberg adjustment for multiple comparisons.

Biological significance

Differences could suggest distinct histories of TE accumulation or persistence. Interpretation must also consider differences in mutation rates, genome assemblies, and repeat annotation.

Methods

Step 1: Data Import
Imported *T. spiralis* and *T. britovi* TE datasets into R using read_csv()

Step 2: Data Filtering
Excluded non-TE annotations and prepared the datasets for analysis.

Step 3: Outlier Detection
Identified potential outliers in percentage divergence using the IQR method.

Step 4: Descriptive Analysis
Calculated mean percentage divergence for each TE matching class in both species.

Step 5: Comparative Visualization
Generated bar charts to compare mean percentage divergence between the two species.

Step 6: Statistical Analysis
Used Wilcoxon tests to identify differences in percentage-divergence distributions, with Benjamini–Hochberg correction for multiple testing.

Figure 1: Outlier detection *T. spiralis*

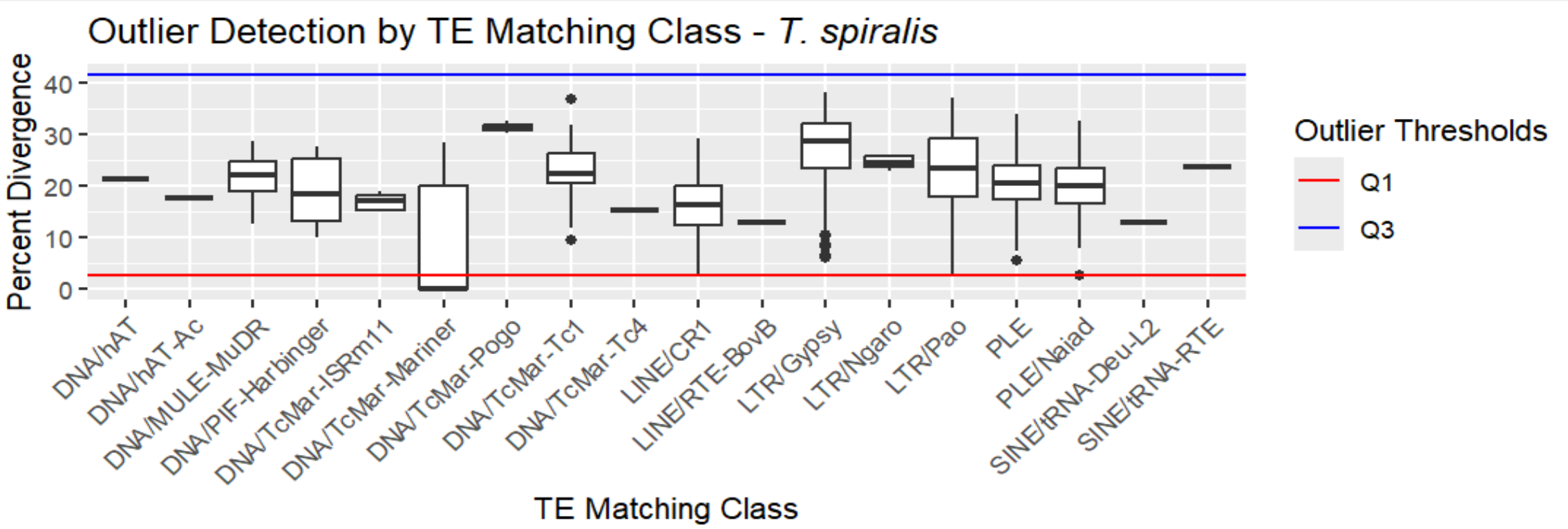


Figure 1. Outlier detection in *T. spiralis* TE annotations.
Boxplots show the distribution of percentage divergence across TE matching classes. Horizontal lines indicate the lower and upper IQR-based outlier thresholds, while individual points beyond the thresholds represent potential outliers.

Figure 2: Outlier detection *T. britovi*

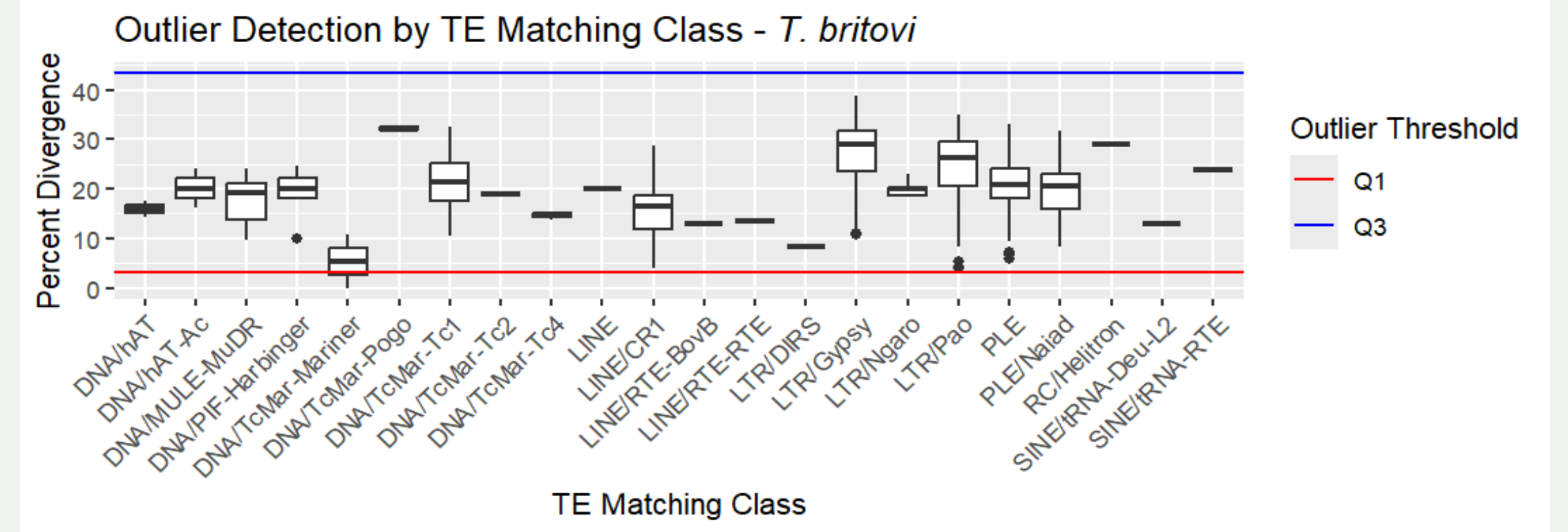


Figure 2. Outlier detection in *T. britovi* TE annotations.
Boxplots show the distribution of percentage divergence across TE matching classes. Horizontal lines indicate the lower and upper IQR-based outlier thresholds, while individual points beyond the thresholds represent potential outliers.

Figure 3: % Divergence in Shared TE Classes of *T. britovi* and *T. spiralis*

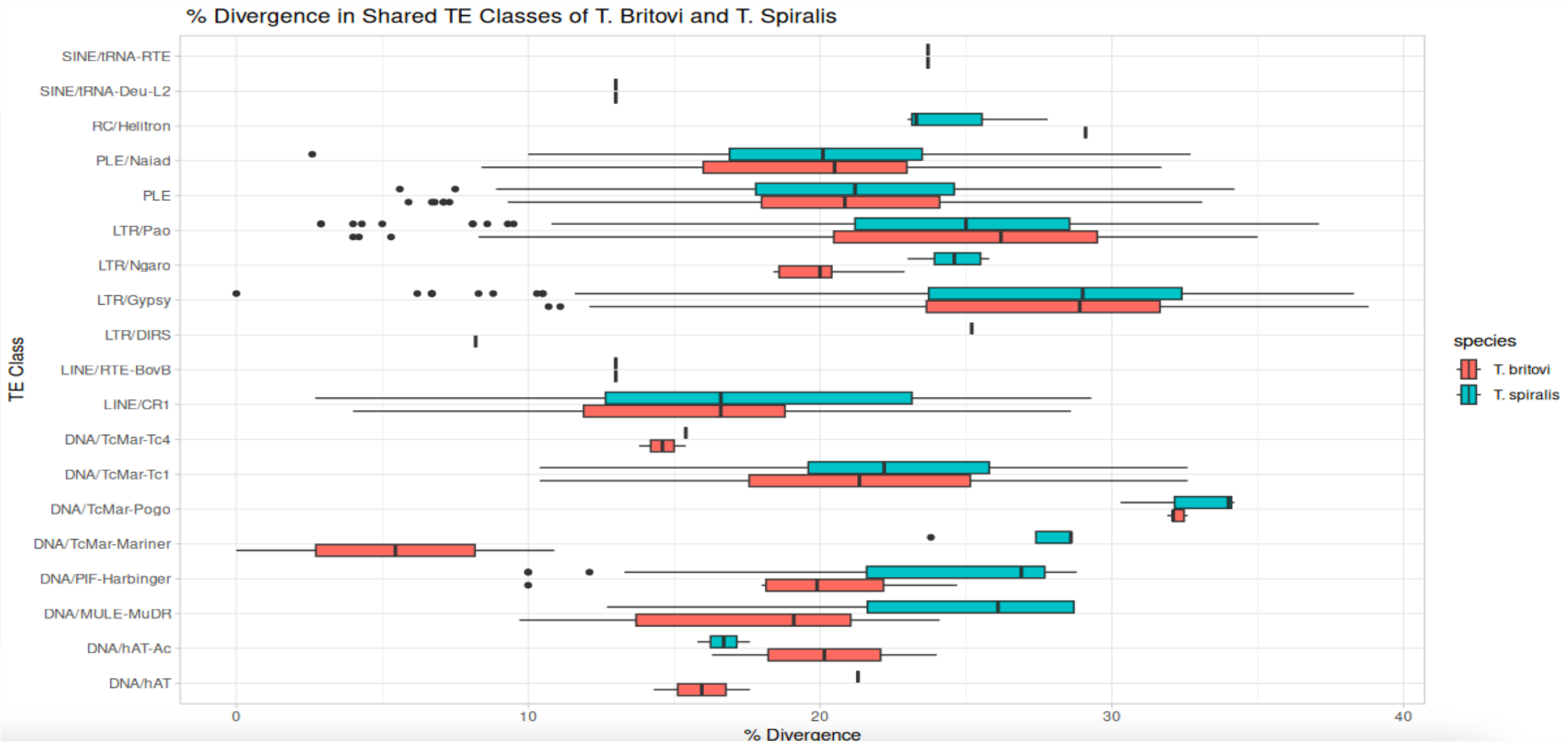


Figure 3. Percentage divergence across shared TE classes in *T. spiralis* and *T. britovi*.
Boxplots compare percentage-divergence distributions between the two species across shared TE classes. The median, spread and individual outlying observations illustrate variation in divergence within each class.

Results + Discussion

- Wilcox test showed no statistical significant differences in % divergence in any shared TE class between *T. britovi* and *T. spiralis*
- Suggests that the two species could have similar histories of TE activity for the shared classes
- % divergence is only a proxy of relative TE age and history, so does not necessarily prove they are identical
- We expected to observe differences based on previous studies done comparing different trichinella species
- However, because TE activity varies widely, results based on different species may not necessarily apply to these ones
- Overall, learned that TE divergence varies widely between classes and that even though they may appear to look different between species on a graph, there needs to be statistical evidence to really support the hypothesis

Strengths & Limitations

Strengths

- Outlier identification: ensures no extraneous values lead to false positive results
- Utilization of Wilcox test: ensures no assumption of normal distribution for percent divergence for all TE classes
- Benjamini–Hochberg adjustment: ensures controlling for multiple tests

Limitations

- Comparison of one sample to another: a single sample might not be representative of the entire species genome average; this sampling artifact may prevent us from catching true differences in TE percent divergence
- Zero counts: to analyze statistical differences between groups, values are needed for a given TE class in both species; it is, therefore, possible that the presence of a TE class in one species but not the other, could reflect historical differences that were statistically undetected

Next Steps & Future Work

Future Work

- Re-perform analysis but with population data (multiple RepeatMasker files for a given species) to elucidate overall changes in TE percent divergence

- Map TEs insertion locations to genome to identify specific genes/biological functions they may alter

References - Coding

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References – Background Information

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World Organisation for Animal Health. (2023). Trichinellosis (infection with Trichinella spp.). In Manual of Diagnostic Tests and Vaccines for Terrestrial Animals (Chapter 3.1.22).

Kimia's Copilot Prompts

- which one of these speasies have been studied most in contrast to Trichinella spiralis relevant to transposable elements: Trichinella nativa Trichinella britovi Trichinella pseudospiralis Trichinella murrelli Trichinella nelsoni Trichinella sp. T8Trichinella sp. T9Trichinella papuae Trichinella zimbabwensis Trichinella patagoniensis
- what type of information could studying TE abundance and distribution provide us with
- Compare Trichinella spiralis and Trichinella britovi in a concise table
- What is the biological rationale for comparing transposable-element sequence-divergence profiles between T. spiralis and T. britovi?
- Explain how TEs differ from simple repeats and low-complexity sequences
- can we learn about the ecological differences in T. Spiralis and T. Britovi from their TE percent divergence
- What statistical test can we use to look for significant differences between perc_div in TE mathcing_class between the two species
- Explain Benjamini–Hochberg adjustment in more details

Kimia's Copilot Prompts

- Summarize findings relevant to our Trichinella TE comparison from these papers:
- Korhonen et al. (2016), “Phylogenomic and biogeographic reconstruction of the Trichinella complex,” DOI: 10.1038/ncomms10513.
- Dayi (2024), “Diversity and evolution of transposable elements in the plant-parasitic nematodes,” DOI: 10.1186/s12864-024-10435-7.

Kriesha's Copilot Prompts

- could transposable elements play a role in increasing reproductive fitness of trichinella species?
- between T. spiralis and T. britovi, does one have higher reproductive capcity/fitness than the other?

Will's Copilot Prompts

- Can you find important papers related to T. britovi
- how is history and % divergence of TEs related
- What is the best graph to compare the distribution of %divergence across 2 different trichinella genotypes
- whats the best statistical test to do if i want to compare the distribution of %div in classes of TEs in trichinella britovi and spiralis
- What are the non-shared classes between britovi and spiralis

Yemi's Copilot Prompts

- Can you find me some literatures with extensive insights into trichinella and their evolutions over the years
- "Give me a brief overview of Trichinella, the essential biological context and not a detailed disease epidemiolgy
- "Find me common images of trichinella"
- "Give me a side-by-side comparison of T.spiralis and T.britovi: their relationship, relevant host associations, and geographic distributions, supported by references."
- "Are there biological rationale for comparing these two species of Trichinella (spiralis and britovi)?"
- "I want to prepare some background to give my reader or audience some bit of definitions and vocabularies. What are transposable elements? Define TE, explain broad categories, and distinguish TEs from other repetitive sequences."

Emmanuella's Copilot Prompts

- How best can I structure the methods section for this assignment? With a table or?
- Can you shorten and correct the grammar for each method in my draft? (file attached)
- Here is a draft of my self-reflection assignment, can you crosscheck it for grammatical and spelling errors, without altering the content itself